



FLEET & SHORE PERSONNEL TRAINING

IAMSAR Training

Search and rescue organisation, the phases of a SAR operation, and the ship's legal and practical role within the IAMSAR system.

IMO IAMSAR Manual

SOLAS Ch. V Reg. 33

Company SMS

What We Will Cover

01

SAR Fundamentals

Legal duty to assist and the structure of the IAMSAR Manual

02

SAR Organisation

RCCs, MRCCs, the SMC, OSC and on-scene units

03

Phases of a SAR Operation

Awareness through mission conclusion

04

Distress Alerting & Communications

GMDSS equipment and the alerting chain

05

Search Planning & Patterns

Datum, drift and choosing a search pattern

06

On-Scene Rescue & the Master's Role

Recovery, survivor care and reporting

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Helicopter Operations

Winching, hi-line transfers and deck safety

08

SAR Action Cards

IAMSAR's ready-reference checklists and message formats



SECTION 1

SAR Fundamentals

01

Why every ship, everywhere, carries a legal and moral duty to respond to distress at sea.

The Duty to Render Assistance

SOLAS and UNCLOS create a near-absolute obligation on every master, reinforced by the IAMSAR Manual's guidance on how that duty is carried out.

- SOLAS Chapter V, Regulation 33 requires the master of a ship at sea to proceed with all speed to assist persons in distress, if in a position to do so.
- UNCLOS Article 98 places the same duty on every flag state: to require masters to render assistance to any person found at sea in danger of being lost.
- The obligation applies regardless of the nationality or status of the persons in distress, or the circumstances in which they are found.
- A master may be released from the duty only on being informed that other ships are already proceeding, or where the ship's own safety would be seriously endangered.
- Failure to assist without valid cause can expose the master, the Company and the flag state to serious legal consequences.



Master's Judgement

The master alone decides how far the ship can safely assist, weighing the persons in distress against the safety of the ship and its own crew — and that decision is recorded in the ship's log.

Three Volumes, One System

The IMO/ICAO International Aeronautical and Maritime Search and Rescue Manual is published in three complementary volumes.

I

Organization and Management — establishing and improving national and regional SAR systems

II

Mission Co-ordination — guidance for planning and coordinating SAR operations and exercises

III

Mobile Facilities — carried aboard ships and aircraft to assist in performing a SAR function

KEY DEFINITIONS

Terms Every Watchkeeper Should Know

The vocabulary used throughout IAMSAR and in every SAR communication.



SRR — Search and Rescue Region

An area of defined dimensions associated with an RCC, within which SAR services are provided.



SMC — SAR Mission Co-ordinator

The official temporarily assigned to coordinate a specific SAR operation.



OSC — On-Scene Co-ordinator

A unit designated to coordinate SAR operations in a specific area on scene.



SRU — SAR Unit

A resource such as a ship, aircraft or shore team, crewed to conduct SAR operations.



SECTION 2

SAR Organisation

How national authorities, rescue coordination centres and ships at sea come together in a single chain of command.

02

RCCs, MRCCs and the SMC

Every coastal state maintains rescue coordination centres responsible for its search and rescue region.

- A Rescue Co-ordination Centre (RCC) is the unit responsible for promoting efficient organisation of SAR services and for coordinating operations within an SRR.
- A Maritime RCC (MRCC) handles maritime cases specifically; many states also operate a Joint RCC covering both aeronautical and maritime incidents.
- On receiving a distress alert, the RCC designates an SAR Mission Co-ordinator (SMC) to plan and coordinate that specific operation until it concludes.
- The SMC gathers information on the distress, identifies SRUs and other assisting vessels in the area, and issues search and rescue instructions.
- Rescue Sub-Centres (RSCs) may support an RCC for a defined part of an SRR, and Search and Rescue Points of Contact (SPOCs) handle cases from non-SOLAS vessels.



First Contact

Any distress heard or received by a ship should be reported to the nearest RCC/MRCC without delay, even if the ship does not intend to proceed — timely information saves lives.

WHO DOES WHAT

Roles On and Around the Scene

As more units arrive, a clear division of responsibility keeps the operation coordinated.



On-Scene Co-ordinator (OSC)

Often the first capable ship on scene; coordinates other units' search patterns, safety and communications until relieved or the SMC takes direct control.



Co-ordinator Surface Search (CSS)

Designated when many surface units are involved, to coordinate search activity specifically among surface vessels.



SAR Unit (SRU)

Any ship, aircraft or ground team tasked by the SMC/OSC to search, rescue or otherwise assist.



Assisting / Merchant Vessel

A vessel not formally tasked but responding to a distress under the master's own duty to render assistance.

COMMUNICATION FLOW

The SAR Chain of Command

Information should flow up and down this chain throughout an operation.

Level	Typical Unit	Primary Responsibility
National	RCC / MRCC	Overall coordination of the SRR; designates the SMC
Mission	SAR Mission Co-ordinator (SMC)	Plans and directs the specific operation
On-Scene	On-Scene Co-ordinator (OSC) / CSS	Coordinates units actually at the search area
Unit	Individual SRU / assisting ship	Executes assigned search pattern or rescue task



SECTION 3

Phases of a SAR Operation

03

From the first sign of trouble to the moment the case is formally closed, a SAR case moves through five defined phases.

BY THE NUMBERS

Five Operational Phases

IAMSAR structures every case, however large or small, around the same sequence of phases.

01

Awareness — a potential emergency is first noted

02

Initial Action — the report is evaluated and action begun

03

Planning — the search area and pattern are worked out

04

Operations — SRUs search, rescue and recover

05

Conclusion — the case is closed and reported

Awareness and Initial Action

What the RCC and the SMC do in the first minutes after a distress is suspected.

- Awareness Phase: a distress alert, an overdue-vessel report, or an uncertainty (INCERFA) is received and logged by the RCC.
- Initial Action begins immediately: the RCC evaluates the information, classifies the emergency phase (Uncertainty, Alert or Distress) and alerts SAR facilities.
- Nearby ships may be requested to maintain a listening watch, alter course toward the reported position, or proceed directly to assist.
- Even before an SMC is formally designated, the first actions taken by any responding ship — logging position, time and communications — materially affect the outcome.
- Accurate, prompt initial reporting from the scene is often the single greatest factor in a fast, successful search.



Log Everything

Time of first sighting or alert, ship's position, course and speed, and every transmission made or received should be logged as the case unfolds — this record supports the SMC's planning and any later investigation.

Planning, Operations and Conclusion

How the case is worked once units are committed, and how it is formally closed.

- Planning Phase: the SMC calculates datum, defines the search area, selects a search pattern and tasks available SRUs — covered further in Section 05.
- Operations Phase: SRUs proceed to the area, execute their assigned patterns, and report sightings, negative results and fuel/endurance status back to the SMC or OSC.
- The OSC coordinates on-scene activity in real time — adjusting patterns, reassigning units, and managing safety as conditions change.
- Conclusion Phase: the case ends either successfully (survivors recovered), or is suspended/terminated by the RCC when further search is judged to have no reasonable chance of success.
- A final SITREP and full documentation are submitted; lessons learned feed back into national SAR plans and, where relevant, the Company's own SMS.



SECTION 4

Distress Alerting & Communications

04

The equipment and procedures that turn a ship in trouble into a coordinated rescue.

How a Distress Alert Reaches an RCC

The Global Maritime Distress and Safety System links a ship's distress alert to shore authorities and nearby vessels.



EPIRB

Emergency Position-Indicating Radio Beacon; float-free or manually activated, alerts via the Cospas-Sarsat satellite system with the vessel's identity and position.



DSC — Digital Selective Calling

A VHF/MF/HF distress button that transmits ship ID, position and nature of distress automatically to shore stations and ships in range.



SART / AIS-SART

Search and Rescue Transponder; once triggered, produces a radar or AIS return that guides searching units directly to the survival craft.



Inmarsat-C / VHF Ch.16

Satellite distress messaging and voice alerting used to send and receive detailed SITREPs and instructions during the case.

PROCEDURE

Alerting, Relaying and False Alarms

Correct use of GMDSS equipment keeps the alerting chain fast and reliable.

- A distress alert should give position, time, nature of distress and number of persons on board as completely as circumstances allow.
- Ships receiving a distress alert or relay should acknowledge, log the details, and proceed or stand by as instructed by the RCC or nearby OSC.
- VHF Channel 16 (and Ch.70 for DSC) remains the primary short-range distress and calling channel; MF/HF DSC and Inmarsat-C extend the same alerting function offshore.
- An accidentally triggered EPIRB or DSC alert must be cancelled immediately with the appropriate authority — false alerts consume real SAR resources and delay genuine cases.
- AIS-SART and radar SART both greatly improve the final, close-range detection of survival craft, especially in poor visibility or heavy weather.



SITREP Discipline

Regular Situation Reports keep the SMC's picture current — report your position, actions taken, sea state and any sightings even when the news is 'nothing found.'



SECTION 5

Search Planning & Patterns

Turning a last known position into a search area a ship can actually cover.

05

From Datum to Search Area

The SMC's planning rests on a handful of core calculations before any unit is tasked.

- Datum is the single most probable position of the search object, corrected for the time elapsed since the last known position or sighting.
- Total water current (leeway plus surface current/wind effects) is applied to the last known position to project where survivors or wreckage have drifted.
- Search area size is set to give a reasonable Probability of Containment, balanced against the number of units and time available.
- A datum with poor accuracy — an old, uncertain last position — forces a larger search area, more units and more search legs than a fresh, precise fix.
- The SMC continuously refines the area as new information — additional sightings, drift calculations, or negative search results — comes in.



Track Spacing

The distance between adjacent search legs (track spacing) is set from the sweep width of the searching unit and sensor — too wide a spacing risks missing the object entirely.

CHOOSING A PATTERN

Standard IAMSAR Search Patterns

Pattern choice depends on datum accuracy, the number of available units and the type of search object.

Pattern	Best Used When	Units
Expanding Square (SS)	Datum is accurate; small area; single unit	1
Sector Search (VS)	Datum accurate; small area; high probability of detection needed	1
Parallel Track / Creeping Line (PS/CS)	Larger area; datum less certain; multiple units available	1+
Track Line (TS)	Search along the intended route of a missing vessel or aircraft	1+

COVERAGE FACTORS

What Makes a Search Effective

Beyond the pattern itself, several factors determine whether a search actually finds what it is looking for.

SW

Sweep Width — the effective detection range of the searching unit and sensor in prevailing conditions

POD

Probability of Detection — the chance a search will find the object, if it is in the area searched

POC

Probability of Containment — the chance the object is actually within the defined search area



SECTION 6

On-Scene Rescue & the Master's Role

06

First actions on arrival, safe recovery of survivors, and the master's ongoing duty to report.

On Arrival at the Scene

First actions shape the outcome of the whole operation.



Do

- Log the exact time and position on arrival at the search area
- Establish communications with the RCC / OSC and acknowledge instructions
- Post extra lookouts and reduce to a safe, effective search speed
- Prepare rescue boat, pilot ladder, scramble net and recovery gear in advance



Don't

- Don't leave the scene without a release from the OSC or RCC
- Don't manoeuvre close to survivors without first assessing their condition and the sea state
- Don't stop sending position and situation updates once tasked
- Don't delay activating the ship's own MOB / SAR checklists and musters

Getting People Out of the Water Safely

Recovery method depends on the survivors' condition, freeboard, sea state and the equipment on hand.

- A rescue boat or fast rescue craft gives the most control in moderate conditions but takes time to launch and itself carries risk to its crew.
- Pilot ladders, scramble nets and purpose-built recovery platforms allow survivors to board directly if they are able-bodied and the ship's freeboard allows.
- Incapacitated or hypothermic survivors need to be lifted, not asked to climb — a Jason's cradle, davit-mounted basket or similar device should be ready.
- Survivors pulled from cold water can suffer circum-rescue collapse; recover and keep them horizontal wherever practicable, not made to stand or exert themselves.
- First aid, warming and monitoring begin the moment survivors are aboard, following the ship's medical guide and shore medical advice (e.g. via a Telemedical Assistance Service) as needed.

The Master's Ongoing Duty

The master's responsibility does not end once survivors are aboard.

- Report survivor numbers, condition and any casualties to the RCC/OSC promptly, and confirm when the ship is ready to resume its passage or continue assisting.
- Maintain full, accurate log entries throughout: times, positions, communications, actions taken and persons recovered.
- Only depart the scene once formally released by the OSC or RCC — never on the ship's own initiative unless safety requires it.
- Submit a full incident report to the Company on completion, supporting both the flag state's SAR case record and the ship's own SMS review.



Master's Judgement, Revisited

As in Section 01, the master retains the final say on how far the ship can safely go — on-scene decisions are made in real time, always balancing the rescue against the safety of ship and crew.



SECTION 7

Helicopter Operations

Bringing a helicopter safely alongside — or overhead — a ship already engaged in a SAR case.

07

How a Helicopter Works With a Ship

IAMSAR describes several methods for transferring survivors, equipment or personnel between a ship and a helicopter.



Winching (Hoisting)

The helicopter hovers overhead while a winch operator lowers a strop, rescue basket or swimmer directly to the deck or water; the most common method for ships without a helideck.



Hi-Line Technique

A weighted line is lowered first and tended by the ship's crew to steady the load and guide the helicopter, reducing swing in high winds or heavy seas.



Deck Landing

Where a ship has a certified helideck, the helicopter lands directly — the fastest method, but only where deck size, strength and obstructions allow it.



Direct Transfer / Static Discharge

Before touching any lowered cable, strop or basket, it must be allowed to touch the deck or water first to earth its static charge.

PREPARATION

Preparing the Ship for a Helicopter

Careful preparation in the minutes before a helicopter arrives keeps the operation safe for everyone involved.

- Establish direct VHF communication with the helicopter and agree course, speed and the ship's intended manoeuvring plan.
- Steer to place the relative wind fine on the bow, as briefed by the helicopter crew, keeping the ship's motion as steady as possible.
- Clear and secure the operating area: strike all aerals, loose gear and halyards that could foul the winch wire or rotor downwash.
- Brief and station a well-drilled winching party in protective equipment (helmets, goggles, gloves) with a fire party and stretcher standing by.
- Rig clear, unambiguous deck markings or a windsock, and keep all unnecessary personnel well clear of the operating area.



Never Earth the Aircraft

Always let the hoist cable, strop or basket touch the deck or sea before handling it — the helicopter carries a static charge that must discharge first, or a severe shock can result.

During the Winching Operation

The winching party's discipline determines whether the transfer goes smoothly.



Do

- Keep the winching area clear of loose lines, tools and open hatches
- Wear all briefed protective equipment throughout the operation
- Follow the helicopter crew's hand signals and instructions exactly
- Keep the ship's course and speed steady until released by the aircraft



Don't

- Don't touch the hoist hook, strop or basket before it has earthed on the deck
- Don't secure the winch wire to any part of the ship under any circumstances
- Don't allow smoking or naked lights anywhere near the operating area
- Don't manoeuvre or alter course without first warning the helicopter



SECTION 8

SAR Action Cards

Ready-reference checklists every bridge team should have to hand the moment a SAR case begins.

08

What the Action Cards Cover

IAMSAR Volume III (Mobile Facilities) provides a set of standard action cards intended to be kept on the bridge for quick reference during an actual case.



Card 1 — Ship in Distress

Immediate actions for a ship that is itself in distress: alerting, position, abandon-ship preparation and survival craft readiness.



Card 2 — Assisting Ship

Steps for a ship proceeding to assist another vessel in distress: reporting, route, and preparing to receive survivors.



Card 3 — On-Scene Co-ordinator

Guidance for a ship acting as OSC: organising other units, allocating search areas, and reporting to the SMC.



Card 4 — Communications & Reporting

Standard message formats for initial distress acknowledgement, situation reports (SITREP) and final case reports.

Keeping Reports Consistent

IAMSAR defines standard formats so any RCC, aircraft or ship can read a report at a glance.

Format	Purpose	Sent By
Initial Action / Acknowledgement	Confirms receipt of a distress alert and states intended action	Any responding ship or RCC
SITREP	Updates the SMC on position, actions, weather and results as the case develops	OSC, SRUs, ships on scene
SURPIC	Surface picture — the position and movement of all surface units involved in the search	OSC or SMC

Final / Closing Report	Summarises the outcome, lessons learned and closes the case	SMC
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COURSE COMPLETE

Every second — and every well-drilled procedure — counts when a life is at stake.

Company DPA — 24/7 emergency line

Nearest RCC / MRCC — via VHF Ch.16 / DSC

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